

Armaan Mittal

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EXPERIENCE

Software Engineer, Engagely.ai

June 2026 – Present

- Build a production **RAG assistant** that replaced manual **CDSO regulatory lookup** for senior physicians, automating retrieval across a 2,000+ document corpus by **developing and consuming LLM APIs** (Azure OpenAI, Cognitive Search).
- Own **backend services and REST APIs** end-to-end (Python, FastAPI, Celery, Redis, RabbitMQ) on a **multi-tenant SaaS** platform, shipping from design through production deployment for enterprise clients (Cipla, ART).
- Design a **multi-agent orchestration system** with role-specialized agents and tool-calling to automate client workflows; build **Next.js** interfaces for internal tooling.
- Debug and resolve **production incidents** via **root-cause analysis**; build **load testing** (1000 virtual users) and implement **SSO/OIDC** authentication for secure multi-tenant access.

Research Intern (Part-time), Samsung R&D Institute Bangalore

Feb 2025 – Nov 2025

- Built low-latency NLP modules (**gibberish detection**, **PII masking**) for **Bixby** across 12 languages, achieving **93.4%** multilabel accuracy; built an **MCP server and client** to expose the tooling.
- **Deployed and monitored** models in production via **CI pipelines**, **Docker**, and **Kubernetes**; reduced end-to-end response latency by **15%** through **quantization** and cache-aware preprocessing.

Student Researcher, CoEDSAI, TIET Patiala

Jan 2026 – May 2026

- Built a **3D U-Net** pipeline for voxel-wise breast tumour segmentation on MRI volumes (PyTorch, MONAI), achieving **85.94% Dice** using patch-based training and a combined Dice + BCE loss to handle severe class imbalance.
- Co-developed **PRISM-Home** (paper under review, *Expert Systems With Applications*), an interpretable smart-home LLM system with hypothesis scoring, dry-run simulation, and an evidence-ledger verifier; achieved **0.91 grounded-clause precision** across 780 benchmark samples.

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, Java, Go, C++, SQL

Technologies: React, Next.js, Node.js, Fastify, Hono, REST APIs, PostgreSQL, MongoDB, Redis, WebCrypto, PWA, Playwright, Docker, Vercel, Cloudflare Workers, AWS, CI/CD

EDUCATION

Thapar Institute of Engineering and Technology, Patiala

August 2022 – May 2026

- BE in Electronics and Computer Engineering with **minor in Conversational AI**.
- **Coursework:** Operating Systems, Computer Networks, Data Structures & Algorithms, Computer Architecture, Object-Oriented Programming, Database Management Systems, Machine Learning, Generative AI.

PROJECTS

OpeningOS | Full-Stack SaaS Chess Opening-Preparation Platform

openingos.amittal.dev | 2026

- Built a full-stack SaaS platform end-to-end: a local-first PWA **frontend** (Next.js) and a **Fastify/PostgreSQL backend** with **auth, sync, permissions, and billing**.
- Designed a **graph-native data model** for opening repertoires (branches, transpositions, lifecycle states) with an **FSRS-based spaced-repetition scheduler** for practice and review.
- Integrated third-party import pipelines (**Lichess**, **Chess.com**) and **Stockfish/WASM** engine analysis; validated with automated smoke tests, static lint, TypeScript checks, and **Playwright E2E tests**.
- **Technologies:** Next.js, Fastify, PostgreSQL, JWT/OAuth, Stripe, Stockfish/WASM, Playwright, GitHub Actions.

SafeSpace | End-to-End Encrypted Stress-Tracking App

safespace.amittal.dev | 2026

- Built **end-to-end encryption** in the browser with **WebCrypto**: PBKDF2-SHA256 (310k iterations), HKDF and AES-GCM, with the data key wrapped by the password and by a one-time recovery key, so the server never sees plaintext.
- Vault API on **Cloudflare Workers** and **D1** keeps only hashes of derived keys and resists account enumeration.
- Reads heart rate over **Web Bluetooth** and scores four signals with per-signal **logistic regression** models trained on WESAD, matching **ONNX Runtime** outputs in tests.
- **Technologies:** TypeScript, React, WebCrypto, Web Bluetooth, Cloudflare Workers, D1, Python, scikit-learn, Vercel.

ACHIEVEMENTS

- **Accepted research paper (ASCML 2026):** “Scalable Multilingual Content Validation: A Dataset-Free Approach to Gibberish Detection”.
- Selected for **Amazon Summer School 2024** (national AI/ML program) and **National Finalist**, Meta Pragati Hackathon.
- **Head, Competitive Programming Contest** (Saturnalia Fest), led a national-level competition with 300 participants; ranked **362 and 535** of 20,000+ on CodeChef [[armaanmittal](https://www.codechef.com/armaanmittal)].
- Mentored 3K+ students in Data Structures & Algorithms via recorded modules and live sessions.